

Instruction set: These are special functions that a processor can perform such as arithmetic instruction, floating point, bit manipulation, etc. ARM supports a variety of instructions from measly mathematical to the complex vector/SIMD instructions. Until recently, ARM was very much into low-power and miniature cores limited to 32-bit addressing modes. However, the firm recently introduced a new family that supports 64-bit which addresses high-performance computing, but we aren't covering that.

The ARM architecture and family: The ARM architecture is the base for all ARM cores available today. Usually, a company creating custom cores licenses the architecture for fine-tuning/tweaking before it builds its own core. Some of these manufacturers are Apple (A series processors) or Qualcomm (Snapdragon Krait), as mentioned earlier.

Using the architecture and fine-tuning it for special work such as Real Time Processing or general-purpose, ARM creates a **family of processors** that goes under the names Cortex-A, Cortex-R, etc.

The ARM core: By varying the core clock speed, instruction sets and processing elements, ARM produces many cores for the SoC builders. One of the most common that you may have heard of is from the Cortex-A family—the Cortex A9 cores are used in almost all smartphones.

It's totally up to manufacturers to opt for the cores they feel are suitable for their SoC needs.

Since we're now done with the processor jargon, let me lay out a selection criteria covering both hardware and software, which I used to pick the best VFM hardware on the market. Also, keep these points in mind when you shop for a tablet or a smartphone.

- **Screen size:** This is the first thing that comes to mind. Screen size selection is subjective and should be based on your taste. Still, for a smartphone, it's better to opt for a screen that's bigger than 10.16 cm (4"). As for tablets, we will be covering both industry standard 17.78 cm (7") and 25.4 cm (10") devices.
- **Resolution:** Ultra-portables bestowed us with resolutions that were higher than earlier laptop models or even higher than the 56 cm (22") monitor. The best part is that these devices offer the ability to pinch to zoom or tap to zoom, which aids in reading even with a lower-size display. Look for the best screen in terms of PPI (the higher the better); do note that some mobile phones boast of higher screen sizes, but the resolution is not on par, which results in a lackluster viewing experience. Usually, a PPI of above 200 is considered very good.
- **Hardware:** The biggest issue with Android is fragmentation. Thus, to get good performance, you need more than a single horse to drive the system. iOS can run very well with half the processing power and works even smoother—however, in Android, you can have better hardware at a lower price point. The general consensus is to avoid devices with single-core processors. You'll feel lag and jitters during Web browsing, which isn't a good experience.

- **So how many cores?** "The more cores, the better," is generally a myth. Architecture and process nodes boost performance more than stuffing in more cores without enhancements. Phones with quad-core SoCs can drain your battery and heat up your phone more often. Phones with Tegra3 can reach a temperature as high as 65°C under usage which makes them practically very hot to handle. Tablets have more breathing room, so even with a power-hogger temperatures stay low. For example, the performance of the Nvidia Tegra 3, a quad-core, is significantly lower than the Qualcomm dual-core Krait SOC (28 nm counterpart).
- **GPU:** There aren't many options in this segment: ARM (Mali), Qualcomm (Adreno), MediaTek (PowerVR) and Nvidia (GeForce) are the only choices one has. Of these, ARM and Mediatek license their GPUs to vendors, so you'll see those in many phones. If you go for an Adreno or Mali GPU-based system, make sure your device has over 1 GB of RAM, since these co-processors tend to share a large chunk of memory, leaving less for your apps.
- **Updates and custom ROMs:** This is an area that is badly plagued and monopolised by vendors. Even though Android is an open source solution, neither you nor Google can control the updates for the devices. Vendors have complete control over it and that's why Android has stagnated with fragmentation over versions. Most devices below Rs 10,000 hardly receive any updates while devices under Rs 20,000 are put on the hold list and may or may not receive version updates from vendors. If you want a future-proof phone that will receive an update for the next version of Android, look no further than Nexus devices from Google. From what I have seen, Samsung is the top vendor when it comes to providing relentless updates and upgrades for their devices, followed by HTC. The worst are Sony and Motorola, which dropped unteamed devices from their update list for absurd reasons. If you are curious and know how to work around this problem with updates, you can always install third-party ROMs available at XDA-forums. These are highly untested and community built, so be aware that you can ruin your device. Some well-known ROM developer groups are Cyanogen, MIUI, AOKP, etc. Here again, the unfortunate approach of most developers is to target high-end devices for their ROM; so if yours is a low-cost device, it may be left with few or no updates. Still, I can't blame these freelance developers, as they work for nothing yet provide you with compelling solutions (if your device is listed) at no cost. So, before buying a device, do drop by XDA to check how well your device is supported there. I won't be covering 4G or LTE baseband details, since their penetration in India is currently limited and won't make an impact in the next two years at least.